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Comparative life cycle assessment of natural gas and coal-based directly reduced iron (DRI) production: A case study for India

Experience I. Nduagu [a,*], Deepak Yadav [b], Nishant Bhardwaj [c], Sabarish Elango [b], Tirtha Biswas [b], Rangan Banerjee [c], Srinivasan Rajagopalan [a]

[a] Corporate Strategic Research, ExxonMobil Research and Engineering Company, Annandale, New Jersey, USA
[b] Council on Energy, Environment and Water, India
[c] Department of Energy Science and Engineering, Indian Institute of Technology, Bombay, India

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ABSTRACT

The energy-intensive steel industry is responsible for >7% of global greenhouse gas (GHG) emissions, of which the process of ironmaking contributes a large majority. Direct Reduced Iron (DRI) manufacturing, a growing technology in ironmaking, offers the potential to reduce emissions. India, as the largest manufacturer of DRI, is characterized by technological diversity, while simultaneously dominated by coal-based DRI processes. This study develops a comprehensive cradle-to-gate (raw material to iron product) life cycle assessment of three distinct technological pathways for producing DRI: direct coal reduction in a rotary kiln (rotary kiln DRI process), coal gasifier and shaft furnace (coal gasifier DRI process), and reformed natural gas and shaft furnace (NG reformer DRI process). A hybrid life cycle inventory approach is employed, involving process modeling, literature data, and industry consultation to establish baseline process metrics, followed by a sensitivity analysis to address the impact of variabilities and uncertainties. Depending on the supply chain and process assumptions, emissions attribution approach to multi-product systems, and the specific choice of inputs and outputs, the life cycle GHG emissions associated with DRI production range from 1391.1 to 1880.0 $kgCO_2e$/t-DRI for the rotary kiln DRI process, 1565.5 to 1969.1 $kgCO_2e$/t-DRI for the coal gasifier DRI process, and 815.3 to 1160.1 $kgCO_2e$/t-DRI for the NG reformer DRI process. While the research considers India as a case study, the methodology and results can be directionally and broadly applicable to other geographies and process conditions.

1. Introduction

The iron and steel industry accounts for >7% (IEA, 2020) of the global greenhouse gas (GHG) emissions, of which the energy-intensive ironmaking process (crucial to primary steelmaking) contributes the vast majority. Further, the share of primary steelmaking (steel produced from iron ore) in global steel output exceeds 80%, while scrap recycling meets about 20% of worldwide demand (IEA, 2020). The major processes for primary steelmaking are the Blast Furnace (BF) – Basic oxygen furnace (BOF), Direct Reduced Iron (DRI) – Electric Arc Furnace (EAF), and Smelt Reduction – BOF routes. The BF-BOF route is primarily coal-based and has a reported overall emissions intensity of 2.2 tonne (t) CO_2/tonne of crude steel (TCS) (IEA, 2020). DRI is characterized by a portfolio of diverse technology options consisting broadly of coal-based and gas-based processes. The CO_2 emissions associated with the natural gas (NG) DRI-EAF route are 1.4 t-CO_2/TCS, while scrap based steel production is reported to have an emissions intensity of 0.3 t-CO_2/TCS (IEA, 2020) due to the absence of emissions involving the energy-intensive ironmaking step. Recently, Fan and Friedmann (2021) have reviewed various low-carbon technology pathways in iron and steelmaking, citing carbon emission values based on prior literature. Notably, the messages from these macro-level studies do not have the benefit of detailed life cycle assessment (LCA) for various DRI pathways.

India's steel sector is unique in technology diversity - integrated (BF-BOF) process co-exists with significant DRI capacity. The Indian ironmaking industry is one of the more GHG-intensive globally, with a noted emissions intensity of about 2.3 t-CO_2/TCS - well above the global average of 1.85 t-CO_2/TCS (IEA, 2020; WSA, 2020a). Heavy reliance on coal, lower-grade iron ore and process inefficiencies are offered as primary reasons for the high emissions intensity. As the second-largest steel-producing country, India produced 110 million tonnes (MT) of steel in 2019 (~6% of global output) (WSA, 2020b). Further, production

* Corresponding author
E-mail address: experience.i.nduagu@exxonmobil.com (E.I. Nduagu).

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capacity is anticipated to rise to 300 MT by 2030 (MoS, 2017). With unavailability of scrap and low per-capita steel consumption levels, a majority of the future steel demand would be met through primary production. As a fast-growing economy with a large base capacity, India offers a unique opportunity to mitigate emissions from primary steelmaking through the adoption of low-carbon technologies and feedstock. This requires an in-depth discussion of the key factors contributing to GHG emissions in various ironmaking technologies. More specifically, a critical analysis, using India as a case study, of emissions-reduction pathways in DRI manufacturing may be especially beneficial, given the significant potential of this technology space towards low-carbon steelmaking technologies of the future.

Material and energy flow for several pathways in the iron and steel industry, including BF-BOF (Griffina et al., 2019), DRI-EAF (Dey et al., 2015), hydrogen-based DRI or H-DRI (Vogl et al., 2018) and hydrogen injection (Yilmaz et al., 2017) in blast furnaces (Tang et al., 2021) have been discussed. An accumulative model was developed for the life cycle comparison of energy, resources and waste factors (Li et al., 2002), but the study does not cover GHG emission factors. On sponge iron-making in coal-based rotary kilns, there are studies on process modeling (Gaurav and Khanam, 2016), energy efficiency analysis (Dey et al., 2015), heat loss minimization (Chakrabarti et al., 2016), and benchmarking (CSIR CIMFR, 2015). Jin et al. (2017) investigated the life cycle emissions from the BF-BOF process. Direct CO_2 emissions associated with DRI production using natural gas (NG), coal-derived syngas, and coke oven gas making have been discussed (Sarkar et al., 2018).

To the best of authors' knowledge, there is no rigorous LCA study comparing various coal and natural gas (NG)-based DRI technologies in practice today - coal-based rotary kiln DRI, coal gasification-based DRI and NG-based DRI manufacturing, even though these options are likely to dominate the DRI technological choices of the future, especially for India. This study attempts to fill the existing gap by presenting a comprehensive, systematic, and comparative LCA of the following three routes for DRI production: i) coal-based rotary kiln, ii) coal gasification-based shaft furnace (hereafter referred to as coal gasifier DRI process), and iii) NG-based shaft furnace (hereafter referred to as NG reformer DRI process. The study uses a unique hybrid approach that combines first-principles-based modeling for material and energy balances with data validation through industry expert consultations, and rigorous implementation of Life Cycle Inventory (LCI) from established databases and publications. A detailed GHG inventory is provided, covering all the life cycle stages of DRI production, from upstream processes (i.e., from extraction and processing of raw materials and fuels) to transportation and use of these materials in a DRI plant. It is expected that the insights from applying this consistent, uniform basis to compare the GHG emissions of the above-mentioned three routes for producing DRI will spur further research interest in low-carbon DRI technologies (e.g., H-DRI) and enable informed policy decisions regarding future trajectories for mitigating emissions from the global steelmaking industry in general and the Indian steel industry in particular.

2. Methodology for life cycle assessment

Recognizing DRI as the uniform primary output from all three processes, the sections below discuss system definition, major assumptions and scenarios explored.

2.1. Scope and system boundaries

Representative processes for rotary kiln, coal gasifier, and NG reformer processes for producing DRI are described in Section S2.2 of the Supplementary Material. The functional unit of the study is one tonne of DRI (t-DRI) produced in the plant. As shown in Fig. 1, the study quantifies the "cradle-to-gate" GHG impacts of DRI manufacturing, covering all stages from raw materials (iron ore and dolomite) and fuel extraction (coal and natural gas) to DRI production, following best practices (ISO 14040, 2006; ISO 14044, 2006). The cumulative GHG (CO_2, CH_4, and N_2O) emissions across the DRI value chain are quantified as CO_2-equivalents (CO_2eq) using the 100-year Global Warming Potential (GWP-100) values (IPCC, 2013).

2.2. Base case assumptions and other scenarios

Table 1 lists input data and parameters that are potential sources of uncertainty and variability in the cradle-to-gate emissions assessment of the DRI processes. A selected set of assumptions for the base case and the other scenarios are shown therein.

Based on industry data and literature, appropriate grades of feed and injection coal are assumed for the base case processes (Supplementary Material S4.1). Lump ore is considered for the rotary kiln process (Sarangi and Sarangi, 2011), whereas the shaft furnace-based processes use pellets (Chatterjee, 2014). The base case for the coal gasifier DRI process uses low ash coal for gasification, and a Top Gas Recycle Ratio (TGRR) of 85% (Section S4.2). Using Sensitivity analysis enables testing the effect of parameters listed in Table 1 on the GHG emissions. For the NG reformer DRI process, the base case uses lean gas with a 98.3 vol% methane content and a TGRR of 65% (Supplementary Material S4.3).

Noting that the coal gasifier and rotary kiln DRI processes generate multiple products, attribution of emissions to DRI and other valuable by-products is carried out using ISO 14044 (ISO 14040, 2006; ISO 14044, 2006) recommendations. For the coal gasifier DRI process, allocation of emissions is avoided by using process partitioning–a method that considers only the segments of the process involved in generating the by-products, but excludes the rest of the operations unconnected to their production. However, the rotary kiln process may generate electricity as a co-product, and in this case, the use of allocation is avoided by using

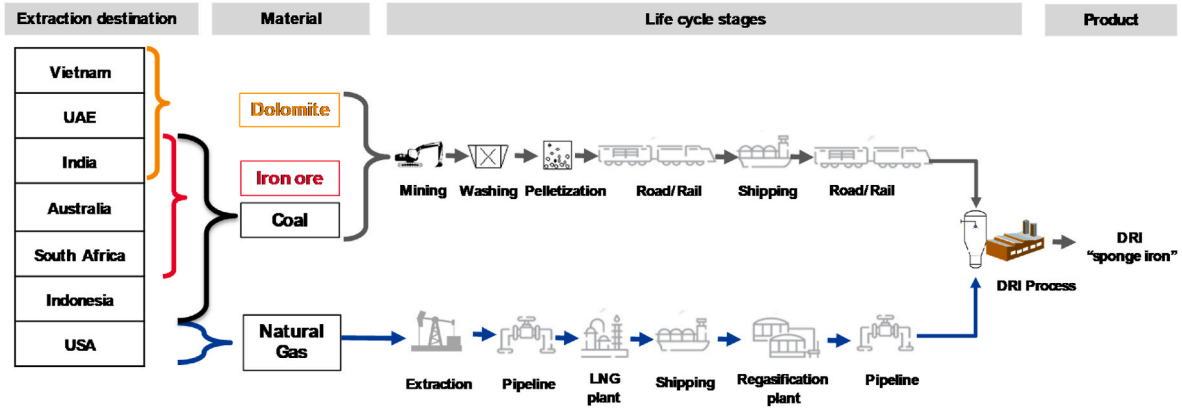


Fig. 1. System boundary for coal-based and NG-based DRI processes.

system expansion, assuming that the generated electricity is exported to the grid.

3. Life cycle inventory

The study uses a combination of relevant literature data and process modeling (using Aspen Plus and Microsoft Excel), validation by consultation with industry experts, and generic data from the Ecoinvent database (Wernet et al., 2016) to generate relevant LCI. Peer-reviewed studies are the first choice of data selection; for example, coal and NG extraction and supply chain data can be reliably sourced from LCA literature. Where complete inventory data is unavailable (e.g., mass, energy and emissions inventory), as is the case for DRI processes, data generated from process modeling is used and validated with inputs from industry experts and other reports. For national grid inventories, Ecoinvent database (obtained via SimaPro software) is used. The GHG inventories for feed and fuel sourcing, industry clusters and transportation pathways are presented below.

3.1. Feed and fuel sourcing

The sponge iron industry in India uses a mix of domestic and imported iron ore, coal and dolomite. The study considers emissions from diesel and electricity used for mining, beneficiation, transportation and shipping for all raw material and fuel. Emissions from coal mining across India (Mallapragada et al., 2019) and other countries are estimated by considering open cast (Mutchek et al., 2016) or underground (Burchart-Korol et al., 2016) mines along with national fuel mix and associated emissions factors (for India, Australia, Indonesia and South Africa). Literature data is used to estimate emissions from iron ore mining in South Africa (Kumbka, 2019), Australia (Haque and Norgate, 2015) and India (IBM, 2014–15, NMDC, 2018). Emissions from ore-to-pellet conversion are also included in the analysis (KIOCL, 2019). Inventory data on dolomite mining are calculated from mine statistics published by the Indian Government (DGMS, 2015). LNG import from the Marcellus shale region in the US (Laurenzi and Jersey, 2013) and associated LCI (Mallapragada et al., 2019) are considered for this study. A detailed description of fuel and feed sourcing can be found in the Supplementary Material (Section S3).

3.2. Industry cluster and transportation mapping

Steel industry clusters in six states are considered (Table 1, Fig. 2), representing approximately 60% of India's total installed DRI capacity in 2019. Distances between the steel plant and the nearest raw material/fuel supply locations are estimated based on geo-spatial mapping. The nearest import terminal is identified from the import-export dashboard (Ministry of Commerce and Industry, 2020).

Transport emissions are obtained by considering freight movement through trucks (CPCB, 2021; DieselNet, 2020), rail (Indian Railways, 2019), and ships. Road transport is assumed to provide last-mile connectivity in cases with no direct rail connectivity. For shipping emissions, a Bulk Shipping Emissions Estimation tool (HEC, 2018) is used to estimate the specific emission factors for ships running on marine gas oil or heavy fuel oil. The transport calculations and discussions are described in Supplementary Material, S3.5-S3.8.

3.3. Process data and analysis

Table 2 lists the process parameters of the rotary kiln, coal gasifier, and NG reformer DRI processes discussed in this study.

3.3.1. Rotary kiln DRI process data and analysis

Modeling data indicates a specific energy requirement of 21.9 GJ/t-DRI (HHV basis) from coal fed to the kiln (Fig. 2, Fig. S6, Supplementary Material). Heat losses in the kiln include the furnace skin (2.2 GJ/t-DRI), DRI product's sensible heat (4.1 GJ/t-DRI) losses, and the char's sensible heat (0.26 GJ/t-DRI) and chemical energy (1.95 GJ/t-DRI). The kiln off-gas carries 4.1 GJ/t-DRI of energy content, which is used for producing 451 kWh/t-DRI electricity in a waste heat recovery (WHR) unit. About 100 kWh/t-DRI is consumed at the plant for meeting the auxiliary load, whereas the residual 351 kWh/t-DRI is assumed to be exported to the grid. The remaining input energy is used to drive various chemical reactions in the kiln. More discussion on process energetics is reported in the Supplementary Material (Section S5.8).

3.3.2. Coal gasifier DRI process data and analysis

The feed coal requirements are a function of the coal quality and quantity (Table 3). For the base case using low-ash coal with an 85%

Table 1
Base case assumptions and other scenarios explored in the analysis.

Sensitivity parameter	Base Case	Other scenarios				
Coal sourcing	Domestic - Talcher	Indonesia – South Kalimantan	Australia – Bowen Basin	South Africa – Ecca Basin	Domestic – Urub, Libara, Bhupalpally, Khottadih, Basundhara	
Coal type[a]	Low-ash	High-ash				
NG type[d]	Lean NG	Rich NG				
TG H_2O scrubbing[d]	50% H_2O Scrubbing	90% H_2O Scrubbing				
TGRR[a]	85%/65%	65%				
Coal washing[b]	76% of coal	66% of coal	No washing			
Industry cluster	Odisha – Cuttack	Jharkhand - Jamshedpur	Chhattisgarh - Raigarh	Karnataka - Bellary	West Bengal - Durgapur	Odisha - Sundergarh
Domestic transport	Road & Rail	Road only				
Iron ore sourcing	Domestic - Keonjhar	Australia - Pilbara	South Africa - Kathu	Domestic – Meghahatuburu, Appalapur, Noamundi, Rourkela,		
Dolomite sourcing	Domestic - Sundergarh	Vietnam – Luc Yen	UAE – Ras al Khaimah	Domestic – Tulsidamar, Baradwar, Bagalkot, Ramgarh, Birmitrapur,		
Attributional methods[c]	Process partitioning and Substitution	Mass allocation	No Attribution			

[a] 85% TGRR and 65% TGRR are the base cases the coal gasifier and NG reformer DRI processes, respectively. An additional case of 65% TGRR for the coal gasifier DRI process is explored.

[b] Indicates the fraction of coal (as mined) washed, which is a function of the coal quality. About 76 wt% and 66 wt% of the coal, respectively, is used for the coal gasifier and coal kiln DRI processes.

[c] Coal gasifier and coal kiln DRI processes generate multi-products that the processes' life cycle burdens are attributed to. Process partitioning is applied to the coal gasifier DRI process, whereas Substitution is used in the rotary kiln process.

[d] Applies to NG-based processes only.

TGRR, about 530 kg/t-DRI (11.64 GJ HHV/t-DRI) of coal is fed to the gasifier. However, switching from low-ash to high-ash coal increases the feed coal requirement by 45% (~240 kg). Oxygen requirement for gasification is 246 kg/t-DRI for the base case–an oxygen-to-coal ratio of 0.47. Switching to high-ash coal increases this requirement by 10%. However, the specific energy and net electricity requirements for the coal gasifier DRI process vary only slightly with coal quality (Fig. 3).

3.3.3. NG reformer DRI process data and analysis

The NG requirements for iron ore reduction vary slightly with composition (Table 3). In the base case, 180.1 kg/t-DRI (11.06 kmol/t-DRI or 265.4 Nm^3/t-DRI) of lean gas is required to generate syngas. In the alternative rich gas case, an additional 1.3 kg NG is needed. A suitable TGRR must be chosen to generate an acceptable syngas quality (H_2/CO ratio >1.6) (Chatterjee, 2014) required for iron ore reduction and to supply the thermal energy needs. The total specific energy requirement for the lean gas case is 10.98 GJ HHV/t-DRI (reduced by 0.1% upon switching to rich gas), showing that the energetics of the process may not be significantly affected by NG composition (Fig. 3).

3.3.4. Data validation

For the rotary kiln DRI process, overall findings are validated with industrial survey data and performance indicators (Agrawal et al., 2016). The base-case energy intensity of 21.9 GJ/t-DRI is near the range of 22.6–23.4 GJ/t-DRI reported by Agrawal et al. (2016). The estimate (100 kWh/t-DRI) of power consumption in the rotary kiln DRI process is also in agreement with industrial data (85–100 kWh/t-DRI) (Agrawal et al., 2016). For power generation using waste heat recovery systems, the estimate of 351 kWh/t-DRI are within the range of best available technology (Worrel et al., 2008) and industry data (Agrawal et al., 2016).

The coal gasifier DRI process model is validated with literature data (Cheeley, 1999; Srivastava et al., 2017) and operational insights derived from industry consultation (Private Communications, 2020). Process data for the base case show oxygen-to-gasifier feed coal ratio of 0.47, similar to the value reported by Yang et al. (2020) for a Lurgi gasifier using lignite coal. This study yields a steam-to-feed coal ratio of 0.99–1.35 - within the range reported by Yang et al. (2020) and Chatterjee (2014). The crude syngas composition (H_2/CO ratio = 1.63) and volume from the model (Table S35, Section S5.8, Supplementary Material) align with previously reported industrial (Chatterjee, 2014; Srivastava et al., 2017) and modeling (Yang et al., 2020) data. Net coal requirement of 830 kg/t-DRI (with a total ash content of 33 wt%) for the base case is close to the value of 840 kg/t-DRI reported for the Sasol-Lurgi gasifier (Chatterjee, 2014).

NG requirement of 180.1 kg/t-DRI (11.06 kmol or 265.4 Nm^3/t-DRI) used for the base case NG reformer DRI process is similar to the previously reported values of 266.67 Nm^3/t-DRI (Fan and Friedmann, 2021),

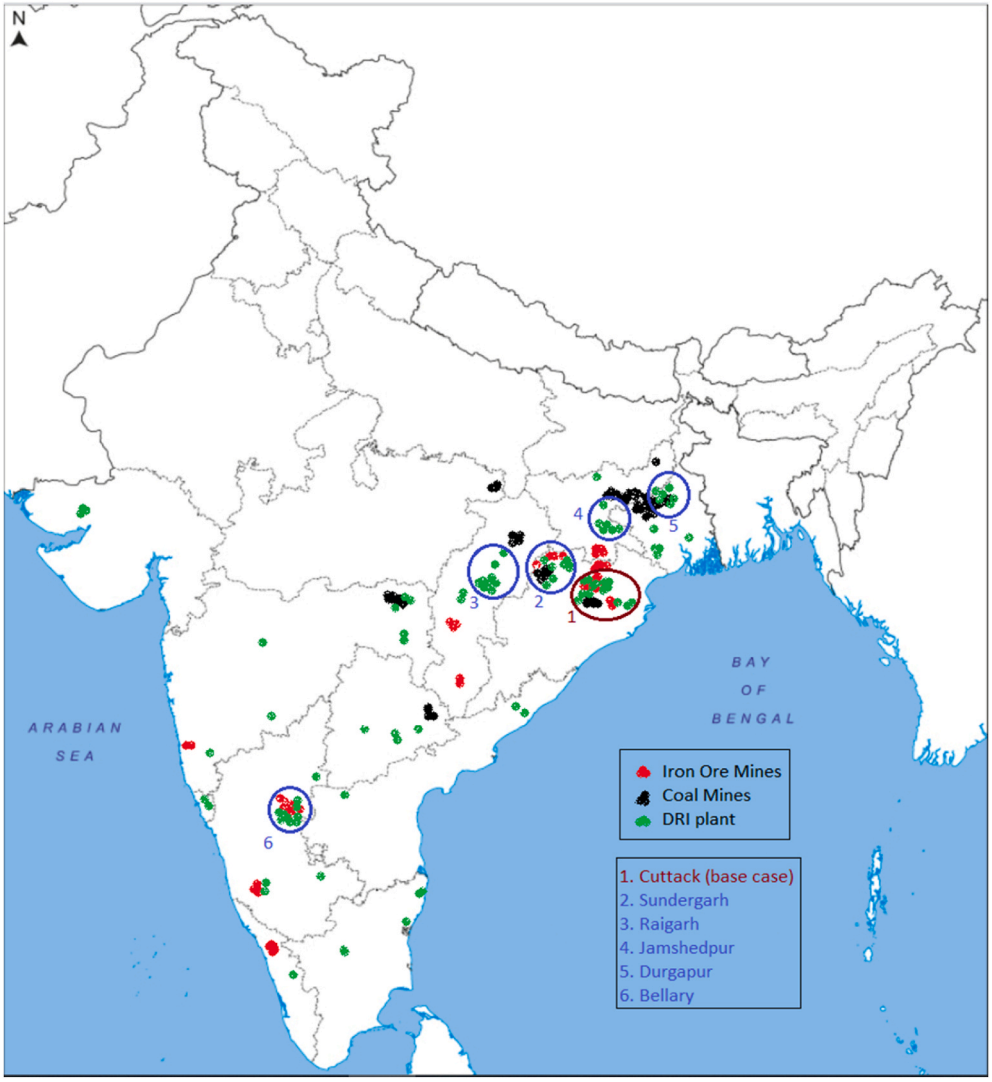


Fig. 2. Map of mainland India showing iron ore mines, coal mines, coal based DRI plants and DRI clusters selected for the life cycle analysis.

Table 2
Inputs, outputs and performance parameters for coal gasifier and NG reformer DRI processes.

Process parameter	Rotary kiln DRI	Coal gasifier DRI		NG reformer DRI	
	Rotary kiln coal	Low-ash coal	High-ash coal	Lean gas	Rich gas
Rotary kiln, gasifier or reformer unit					
Input raw material					
Natural gas (kg/t-DRI)	–	–	–	198.9	200
Feed coal (kg/t-DRI)	611	530	770	–	–
Fuel coal (kg/t-DRI)[a]	465	300	310	–	–
Oxygen/Air (kg/t-DRI)	3228	246	270	–	–
Steam (kg/t-DRI)	–	717	759	–	–
Dolomite (kg/t-DRI)	50				
Shaft furnace and RK process					
Input raw materials					
Iron ore pellets (kg/t-DRI)	1574	1375	1375	1375	1375
Oxygen (kg/t-DRI)	–	42	42	–	–
Clean syngas[b]					
Mass flow (kg/t-DRI)	–	337.9	336.6	975.0	983.8
Volume flow (Nm^3/t-DRI)	–	618.5	617.1	1846	1798.9
H_2/CO ratio	–	1.63	1.63	1.67	1.6
Output materials	–				
TG/Kiln off-gas mass flow (kg)	4250	1103.3	1101.6	1247.7	1262.8
Char (kg)	157				
Flue dust (kg)	256				
Tar		13.3	19.3		
Gas oil		8.5	12.3		
Phenolic pitch		2.2	2.2		
Electricity Requirements					
ASU Power required (kWh/t-DRI)[c]	–	80.6	87.4	–	–
Power required (kWh/t-DRI)[d]	100	320.6	327.4	120	120
Power produced (kWh/t-DRI)	451	29	29	–	–
Net power needed (kWh/t-DRI)	−351	291.6	298.4	120	120

[a] Fuel coal, combusted together with TG for steam generation and feed preheating, has the same composition as high-ash coal.
[b] Produced after syngas scrubbing, which is only applicable to the coal gasifier process. The composition of the syngas from the NG reformer process is unchanged because it is used directly in the shaft furnace without syngas cleaning.
[c] Not modeled but assumes 280 kWh/tonne O_2 as the specific power requirement for oxygen production (Air Liquide, 2017).
[d] Includes power requirement for ASU, which is only applicable to the coal gasifier process. The electricity requirement for NG reformer DRI process is from Chatterjee (2014).

and 11.15 kmol/t-DRI (Sarkar et al., 2018), but with unspecified NG composition. An additional 18.8 kg NG is needed to compensate for heat losses, bringing the total NG requirement to 198.9 kg/t-DRI. These process energy losses do not appear to be captured in previous studies (Fan and Friedmann, 2021; Sarkar et al., 2018). The NG requirement translates directly to the specific energy of 9.88 GJ (Low Heating Value)/t-DRI or 10.98 GJ HHV/t-DRI, which is within the range (9.5–10.5 GJ LHV/t-DRI) reported elsewhere (Chatterjee, 2014). Syngas and TG composition, and the TGRR utilized in this study correspond to that reported elsewhere (Sarkar et al., 2018; Chatterjee, 2014). A TGRR of 65% was determined to be an appropriate design choice to generate acceptable syngas quality (Chatterjee, 2014) for iron ore reduction.

Table 3
Upstream GHG emissions (kg/t-DRI) from Rotary kiln, Coal gasifier DRI and NG Reformer DRI processes.

Sr. No.	Component	Rotary kiln DRI	Coal gasifier DRI	NG Reformer DRI
1	Coal mining	88.9	70.8	–
2	Coal transport	5.2	4.1	–
3	Iron ore mining	13.3	11.7	11.7
4	Iron ore transport	7.7	6.7	6.7
5	Iron ore concentration	1.4	1.2	1.2
6	Iron ore pelletization	–	252.4	252.4
7	Dolomite mining	0.2	0.2	0.2
8	Dolomite transport	0.4	0.4	0.4
9	NG upstream			181.3
Total		117.0	347.5	453.9

3.4. Criteria pollutants

Upstream and plant criteria emissions of SO_x, NO_x and Particulate matter (PM) are reported based on literature estimates. While assuming that industries adhere to emissions regulation wherever mandated by norms, this study considers uncontrolled criteria emissions for unregulated sectors. Plant-level emissions are calculated by mass balance and actual measurements indicated by UNEP (1983), CPCB (2007) and US EPA (1998).

Unlike GHG emissions that have a global footprint, criteria pollutants have a local impact. Therefore, this study lists unregulated criteria pollutants emitted to the atmosphere at various phases. Details on the methodology and approach are expounded in the Supplementary Material (Section S5.11). In general, criteria pollutant emissions across upstream processes are proportional to the energy use and the respective GHG emissions.

4. Results and discussion

Emissions from upstream processes (4.1) and DRI plant (4.2) are presented first, and then aggregated into the life cycle “cradle-to-gate” results (4.3), following which a detailed discussion on key sensitivity parameters is presented (4.4).

4.1. Upstream process emissions

Upstream emissions for the rotary kiln, coal gasifier and NG DRI processes are provided in Table 3. The largest contributions are from iron ore pelletization, followed by NG supply and coal mining supply chains (for more details, refer to S5.1-S5.7 in the Supplementary Material). The coal gasifier and NG reformer DRI processes use iron ore pellets (67% Fe) compared to the rotary kiln process that uses lump ore (64.1% Fe). Thus, pelletization emissions are not considered in the base case of the rotary kiln process. Due to relatively small amounts of dolomite used, the dolomite supply chain emissions are assumed to be negligible compared to other components.

4.2. DRI process emissions

Fig. 4 shows a breakdown of the plant GHG emissions for the three routes considered in the study. In general, the rotary kiln process has the highest direct GHG emissions, followed by the coal gasifier DRI process and the NG reformer DRI process.

The total GHG emissions from the rotary kiln process are 1763 kg CO_2/t-DRI with ~84% being direct emissions from the kiln, while the rest are from the After Burning Chamber (ABC) unit. Captive electricity generated in the rotary kiln DRI process exceeds the electricity

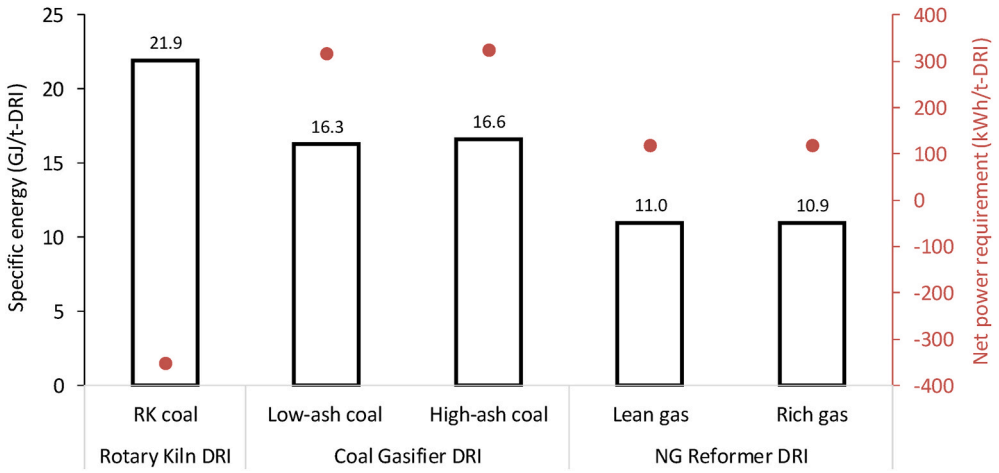


Fig. 3. Specific energy (GJ High Heating Value [HHV], on the primary y-axis) and net electricity requirements (kWh, on the secondary y-axis) of coal-based and NG-based DRI processes.

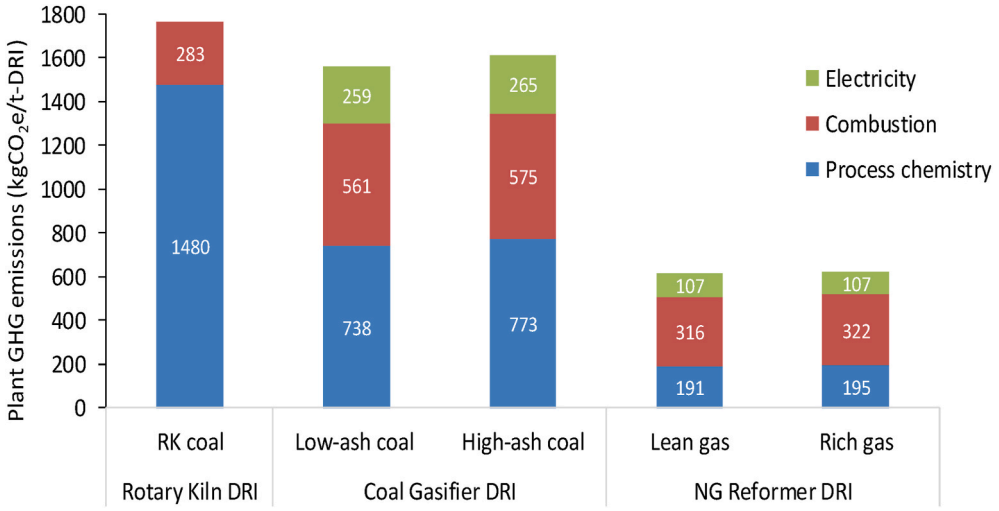


Fig. 4. Breakdown of plant and electricity-based GHG emissions associated with coal-based and NG-based DRI processes.

requirement, allowing for surplus electricity to be exported to the grid. For the coal gasifier DRI route, the process chemistry and combustor emissions make up 46% and 37% of the process GHG emissions, respectively, whereas electricity use emissions comprise the rest (~17%). For the NG reformer DRI process, the direct plant GHG emissions are 507 kgCO_2e/t-DRI, whereas the indirect GHG emissions from electricity use are 107 kgCO_2e/t-DRI. Electricity-related GHG emissions are the same for the lean and rich NG cases, given the equivalent electricity requirement. Similarly, direct plant GHG emissions for the lean gas case are 1.8% (9 kgCO_2e/t-DRI) lower than that of the rich gas case.

4.3. Life cycle emission results

Fig. 5 shows the life cycle GHG emissions intensity of the three DRI routes. The base case for the rotary kiln process has a total life cycle emissions of 1880 kgCO_2e/t-DRI, without accounting for surplus electricity. About 290 kgCO_2e can be attributed to the exported power of 351 kWh/t-DRI based on the amount of export and grid emissions intensity. The resulting net life cycle GHG emissions of the rotary kiln process become 1590 kgCO_2e/t-DRI. Similarly, the gasifier DRI process has a total life cycle emissions of 1905.6 kgCO_2e/t-DRI, without emission attribution to other by-products. With energy-based process partitioning, the net life cycle GHG emissions of the coal gasifier DRI process are 1818 kgCO_2e/t-DRI. Given that the NG reformer DRI process generates no by-products, the entire life cycle GHG emissions (1068 kgCO_2e/t-DRI) are assigned to the DRI product.

Process plant emissions account for the majority of the GHG emissions of the DRI processes, resulting in 93%, 81%, and 58% of the net life cycle GHG intensity of the rotary kiln, coal gasifier, and NG reformer DRI processes, respectively, signifying the greatest opportunity for emissions reduction. Switching from coal to NG as the feedstock leads to a 58% reduction in the GHG emissions of the coal-based DRI process plants (including electricity use). On a life cycle basis, GHG emissions from the NG reformer DRI process are lower than for the coal-based rotary kiln and gasifier DRI processes by 33% and 41%, respectively, when uniformly compared across base case scenarios.

Thermal (i.e. heat-related) plant emissions for the coal gasification process (1222.6 kg CO_2/t-DRI) are lower than for the rotary kiln process (1480 kg CO_2/t-DRI). However, the life cycle emissions from the coal gasifier process are higher than the rotary kiln process due to the electricity required for the plant auxiliary load (251.5 kg CO_2/t-DRI) and the pelletization (252.4 kg CO_2/t-DRI) process applicable to the coal gasifier process. Similarly, the life cycle emissions from NG-based process increase primarily due to the upstream processes for producing and transporting natural gas (181 kg CO_2/t-DRI), ore pelletization (252.4 kg CO_2/t-DRI) and the auxiliary power requirement (106.6 kg CO_2/t-DRI)

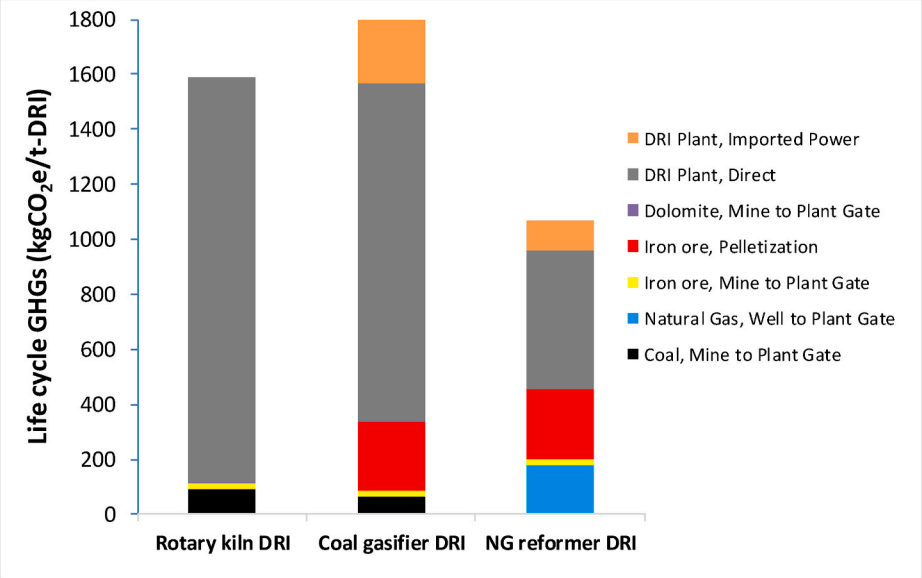


Fig. 5. Life cycle GHG emissions of the coal-based processes (rotary kiln and coal gasifier MIDREX processes) and NG-based MIDREX process for base-case scenarios.

at the DRI plant. Clearly, the hierarchy of technologies differs when viewed from the basis of process-related emissions versus life cycle emissions. These results therefore highlight the importance of uniform development and application of LCA as a tool for quantifying the GHG impacts of DRI fuel, feedstock and technology options on a holistic, cradle-to-gate basis.

4.4. Sensitivity analysis

Noting that the sources of variability in the rotary kiln DRI process (Fig. 6A) span the supply and process chains, the following parameters are most sensitive to the GHG emission intensity of DRI: emissions attribution to the electricity by-product generated via waste heat recovery (18.2%), iron ore pelletization (15.9%), sourcing of coal and iron ore from different geographies (6.7–13.4%), VM-C combustion in the kiln (−12.5%), waste heat recovery efficiency (−6.5%–6%) and injection coal quality (−1.2%–4.7%). Integrating the rotary kiln process with a waste heat recovery unit significantly impacts the life cycle GHG intensity. With no waste heat recovery, the GHG intensity increases to 1880 $kgCO_2e$/t-DRI. Similarly, the captive power unit's net efficiency has a significant impact on the life cycle GHG emissions of the coal kiln process. Using pelletized ores instead of lump ores increases the net life cycle GHGs by ~16%. The life cycle GHG emissions of the rotary kiln DRI process are sensitive to the process modeling uncertainty relating to combustion of the feed coal's volatile matter (VM-C) in the kiln. If the VM-C burns in the kiln, it significantly reduces the plant's feed and injection coal consumption, thus also reducing the net life GHG cycle emissions from 1590 to 1391 kgCO2e/t-DRI. Using domestically sourced raw material provides significant GHG emission benefits relative to sourcing coal, iron ore, or dolomite from abroad.

For the coal gasifier DRI process, the following parameters are the most sensitive to the GHG emission footprint of DRI (Fig. 6B): iron ore pelletization (−13.9%), coal and iron ore sourcing from South Africa, Australia, or Indonesia (4.3%–8.3%) and emission attribution to by-products (4.9%). The GHG emission footprint of the coal gasifier DRI process is highly sensitive to a set of parameters related to sourcing and transport of coal and iron ore, specifically mining, iron ore pelletization, and inter-country shipping and transport. A GHG reduction of 252.4 $kgCO_2e$/t-DRI (~13%) from the base case could be realized if concentrated iron ore lumps can be used in lieu of pelletized ores. Similar to the rotary kiln DRI case, the use of domestically sourced raw materials provides significant GHG emission benefits over imports. The implementation of energy-based process partitioning for multi-product emission allocation leads to a lower-range emissions burden – 95.4% of life cycle emissions of the process – assigned to the DRI product, resulting in a GHG emission footprint of 1817.9 $kgCO_2e$/t-DRI. However, the GHG emission intensity of DRI increases by 2.9% if mass-based emission allocation is applied and by 4.6% if all the life cycle emissions are assigned to the DRI product.

The life cycle GHG footprint of the NG reformer DRI is sensitive to iron ore sourcing and pelletization, upstream NG production, a mix of NG composition and the extent of TG water scrubbing, and inland transport mode (Fig. 6c). Similar to the trend observed with the coal-based DRI processes, iron ore pelletization is the most sensitive parameter, contributing to about 252 $kgCO_2e$/t-DRI or 24% of the life cycle GHG footprint.

Related to upstream emissions, a recent study (Rutherford et al., 2021) reported higher methane emissions from the oil and gas sector in the US than those reported by the U.S. Environmental Protection Agency GHG inventory (EPA, 2020) in 2015. They attributed the divergence in emissions to a few sites referred to as super-emitters experiencing episodic events under abnormal operating conditions, such as in liquid storage tank hatches and vents (Alvarez et al., 2018). Given the nature of the equipment from which the higher methane emissions are potentially released, results by Rutherford et al. (2021) may be attributable to the crude oil production boundary in a few regions dominated by these so-called super-emitters. The results do not affect the GHG footprint of natural gas production because they lie outside the system boundary of natural gas systems. On the other hand, Mallapragada et al. (2019) used relevant data of natural gas systems obtained from field measurements (Mitchell et al., 2015; Zimmerle et al., 2015). Additionally, the data aligns with a recent aerial survey data of the Permian Basin (Contreras et al., 2021) where fugitive methane emissions from compressor stations are similar in magnitude to previous estimates by Mitchell et al. (2015).

Using the 10th and 90th percentile of GHG emission values for upstream NG (Laurenzi and Jersey, 2013) as the range for sensitivity analysis, the variability in the GHG emissions associated with NG production could materially affect the life cycle GHG impacts of DRI production. The life cycle GHG emissions of the base case for NG reformer DRI (1067.6 $kgCO_2e$/t-DRI) can be reduced to 1045 $kgCO_2e$/t-DRI (2.1% reduction) or increased to 1130 $kgCO_2e$/t-DRI (5.8% increase) using the 10th or 90th percentile upstream NG emission values. Other

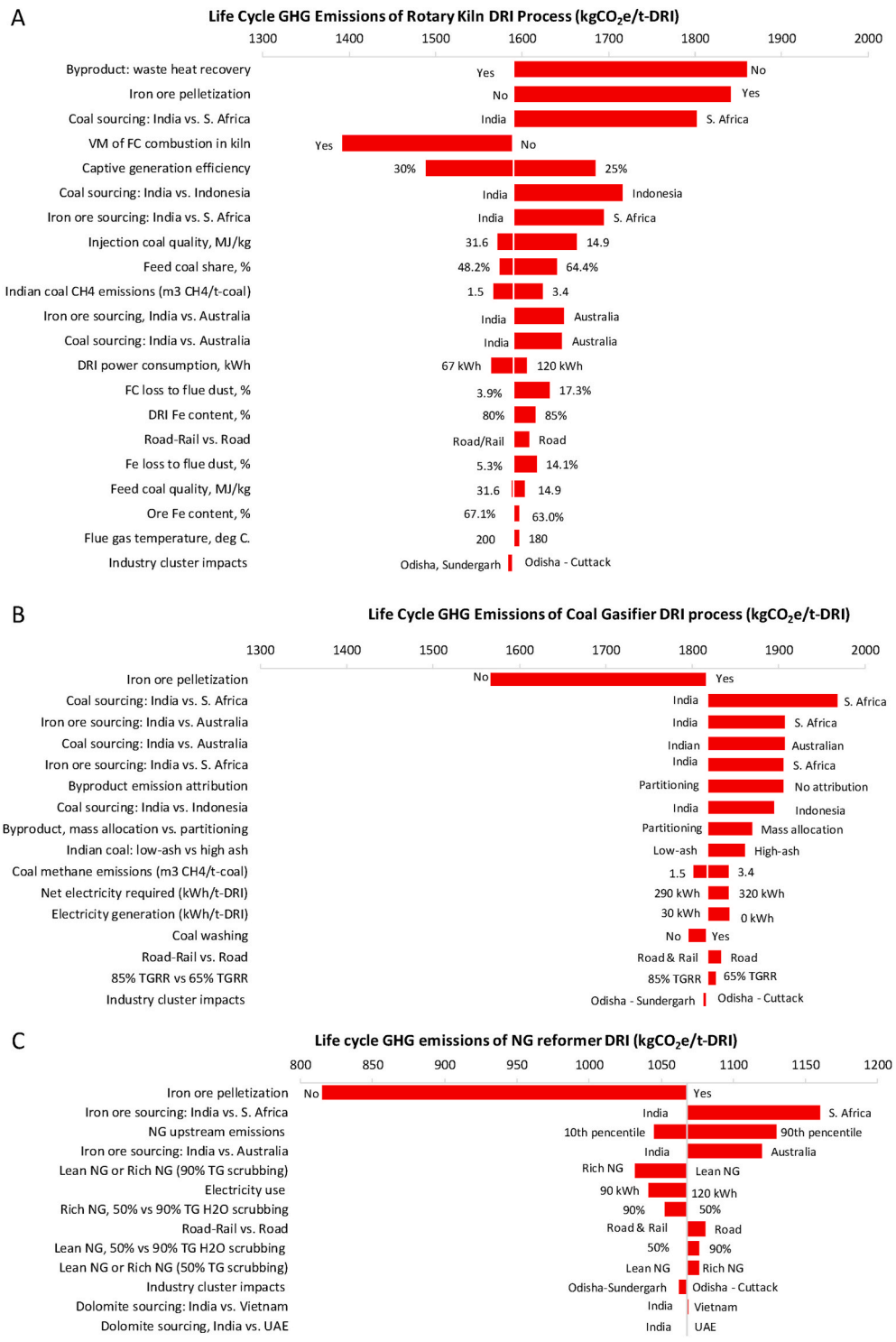


Fig. 6. Sensitivity analysis showing the most important life cycle parameters for: A) Rotary kiln DRI process, B) Coal gasifier DRI process and C) NG reformer DRI process.

parameters such as process electricity use, inland transport mode (rail plus road vs. exclusively road transport) in India, NG composition, TG water scrubbing, and electricity use have a relatively modest effect on the DRI production's life cycle GHG emissions.

5. Conclusions

This study uses life cycle inventory from process modeling, literature, and data from industry consultation to provide a consistent, uniform and comprehensive comparison of the GHG emissions and criteria pollutant impacts from rotary kiln, coal gasifier and NG reformer processes for unit production of DRI. The comprehensive assessment presented here highlights the significance of feedstock/fuel-switching and technology options in reducing energy requirements and GHG emissions associated with DRI production. Replacing coal-based (rotary kiln and coal gasifier) DRI processes with NG-based process reduces the life cycle GHG footprint of DRI by ~33–41% in the analyzed base cases. The life cycle GHG emissions of the coal gasifier and the rotary kiln DRI

processes are noted to be similar.

Depending on the supply chain and process assumptions, emission attribution to by-products, and the specific choice of inputs and outputs, the life cycle GHG emissions associated with DRI production range from 1391 to 1880 $kgCO_2e$/t-DRI for the rotary kiln DRI process, 1566 to 1969 $kgCO_2e$/t-DRI for the coal gasifier DRI process, and 815 to 1160 $kgCO_2e$/t-DRI for the NG reformer DRI process. With the range of uncertainties and variabilities considered, sensitivity analysis results show that the NG reformer DRI process has the potential to lower life cycle GHG footprint of DRI from the coal-based processes by 17–59%. The rotary kiln DRI process offers 11% and 4.5% lower GHG emissions on the lowest and highest end of the sensitivity band, when compared to the coal gasifier DRI process. However, if ore pelletization is uniformly considered for all DRI process, the emissions of the coal gasifier process could be up to 7.7% lower than the rotary kiln process (higher range). Therefore, the choice of operational conditions and supply chain parameters determines which coal-based DRI has the lowest GHG footprint.

Even though the process emissions are lower for the coal gasification process than the rotary kiln, the life cycle emissions from the former are higher than the latter due to the imported electricity and iron ore pelletization requirements. This underscores the importance of uniform, consistent approaches and application of LCA as a tool for quantifying the GHG impacts of different fuel and technology options on a holistic cradle-to-gate basis.

The study can be further extended to refine the process energetics by considering reaction kinetics aspects of the three routes considered in the analysis. Further, a comparative techno-economic analysis can be done to compare the cost-competitiveness of the three processes considered for the analysis. Approaches in this study can also provide a consistent and uniform basis to assess the emissions reduction potential of Hydrogen-based DRI (H-DRI) processes, projected to be a long-term, deep GHG reduction solution for iron and steelmaking. Specifically, H-DRI studies require significant system-level assessment of blue, gray and green hydrogen pathways that merit a separate study. On a life cycle emissions basis, this study establishes that technologies like NG reformer DRI offer significant potential for GHG emissions reduction relative to coal-based processes for DRI manufacture in India. The economic, socio-political, and policy implications of the technology transition to lower-carbon production processes of the future lie outside the scope of this study, and also merit separate discussion.

CRediT authorship contribution statement

Experience I. Nduagu: established the Investigation approach, Methodology, Visualization, writing, Writing – review & editing of the manuscript. **Deepak Yadav:** established the Investigation approach, Methodology, Visualization, writing, Writing – review & editing of the manuscript. **Nishant Bhardwaj:** established the Investigation approach, Methodology, Visualization, writing, Writing – review & editing of the manuscript. **Sabarish Elango:** established the Investigation approach, Methodology, Visualization, writing, Writing – review & editing of the manuscript. **Tirtha Biswas:** Contributed through Conceptualization, Visualization of the study, writing, Writing – review & editing of the manuscript. **Rangan Banerjee:** Contributed through Conceptualization, Visualization of the study, writing, Writing – review & editing of the manuscript. **Srinivasan Rajagopalan:** Contributed through Conceptualization, Visualization of the study, writing, Writing – review & editing of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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